

Isotretinoin: Indications, Mechanism, and Safety Considerations

Indications and Efficacy

Isotretinoin is approved for the treatment of severe, recalcitrant, nodular acne. However, it is frequently used for other forms of significant acne that are unresponsive to conventional therapies, including oral antibiotics, particularly when there is risk of physical or emotional scarring. It is also effective in treating gram-negative folliculitis, pyoderma faciale, and acne fulminans.

One of the most notable features of isotretinoin therapy is the high rate of complete remission, which is typically long-lasting—ranging from months to years—in the majority of patients.

Mechanism of Action

The exact mechanism of action of isotretinoin is not fully understood. The drug profoundly suppresses sebaceous gland activity, which plays a key role in the initial clearance of acne. In some patients, this suppression persists for at least one year, but in most, sebum production returns to baseline within 2 to 4 months after starting treatment.

Side Effects

Due to the widespread expression of retinoic acid receptors (RARs), isotretinoin commonly causes side effects resembling chronic hypervitaminosis A syndrome. These are generally dose-dependent.

Dermatologic and Mucocutaneous Effects

- **Cheilitis** (lip inflammation) occurs in nearly all patients.
- Dry mucous membranes, **xerosis** (dry skin), and **pruritus** (itching) affect over 50% of users.

- Eczematous dermatitis may develop, especially in cold, dry climates.
- Less common effects include hair thinning and granulomatous paronychia.

Ophthalmologic Effects

- Xerophthalmia (dry eyes)
- Night blindness
- Conjunctivitis
- Keratitis
- Optic neuritis
- Corneal opacities have also been reported.

Auditory Effects

Both transient and persistent hearing loss have been associated with isotretinoin use.

Neurologic Effects

Pseudotumor cerebri (benign intracranial hypertension) can occur, presenting with severe headache, nausea, and visual disturbances. The risk is increased when isotretinoin is used concomitantly with tetracyclines; therefore, this combination should be avoided.

Psychiatric Considerations

Potential psychiatric side effects include depression, suicidal ideation, psychosis, and aggressive or violent behavior. However, large population-based studies have not established a causal link. One cohort study comparing isotretinoin users to those on oral antibiotics found no increased risk, with a relative risk of approximately 1.0 for depression or psychosis. Nonetheless, careful psychiatric screening—especially in adolescents, who have a baseline depression risk of 10% to 20%—is recommended.

Gastrointestinal and Hepatic Effects

- Gastrointestinal symptoms such as nausea, esophagitis, gastritis, and colitis are uncommon but possible.
- Acute hepatitis is rare.
- Liver function tests should be monitored regularly, as liver enzyme elevations occur in about 15% of patients, often requiring dose adjustment.

Lipid Abnormalities

- Elevated serum triglycerides occur in approximately 25% of patients, typically within the first 4 weeks of therapy.
- This increase is dose-dependent and often accompanied by a rise in total cholesterol and a decrease in high-density lipoprotein (HDL) levels.

Musculoskeletal Effects

- Myalgias (muscle pain) occur in about 15% of patients.
- In severe cases, creatine phosphokinase (CPK) levels should be checked to rule out rhabdomyolysis.

Teratogenicity and Pregnancy Prevention

The most serious concern with isotretinoin is its potent teratogenicity. The drug is not mutagenic but disrupts fetal organogenesis, leading to severe congenital malformations if taken during pregnancy.

Isotretinoin should be reserved for patients who have failed other treatments. Women of childbearing potential must be fully informed of the risks and must use **two highly effective forms of contraception**, such as an oral contraceptive combined with condoms and spermicidal jelly. Contraception must begin at least **one month before** starting isotretinoin therapy.

Due to these risks, isotretinoin use in the United States is strictly regulated under the **iPledge program**, implemented in March 2006.